A Mathematical Model to Simultaneously Calculate Growth and Death Rates of Cells Undergoing Oxidative Stress

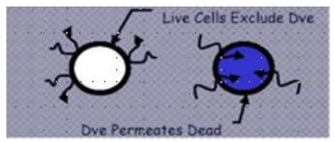
Jackie Penaloza, Ahmad Essway, Ashley Oglesby, Laura LoBuglio, Dr. Meri Firpo

College of Science and Engineering, Department of Mathematics

Abstract

An increasing number of Americans are diagnosed with diabetes mellitus each year. The ultimate goal of this study is to find the best ways to evaluate beta cell proliferation and death in a vitro model of beta cell death due to oxidative stress. The model system used was a mouse beta cell (βTC-6) line and hydrogen peroxide as a source of oxidative radicals to trigger death The hypothesis of this study is that cell growth and death are occurring at the same time and it is necessary to quantify both processes to fully understand the beta cell death in diabetes. Standard trypan blue dye exclusion assays were performed initially and sigmoidal growth curves were established for live and dead cells. However, the number of cells that could be counted was not sufficient to obtain accurate quantification of cell death. For this reason, another exclusion assay with fluorescent 7-Amino-Actinomycin (7-AAD) dye was used with flow cytometry. Flow cytometry, unlike trypan blue staining, collects a larger cell sample, thus increasing accuracy and facilitating both live and dead cell counts. A positive control of ethanol used as a toxin to validate this assay. In addition, the cytoplasmic enzyme lactate dehydrogenase (LDH) was measured in cell culture medium to estimate the number of cells that has been lysed, releasing cytoplasmic contents. Based on these results, a mathematical model can be established to evaluate the quality of the potential cell insults active during the progression of diabetes, and identify novel drug inhibitors.

Trypan Blue Exclusion Assay

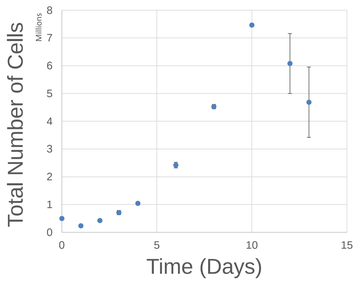


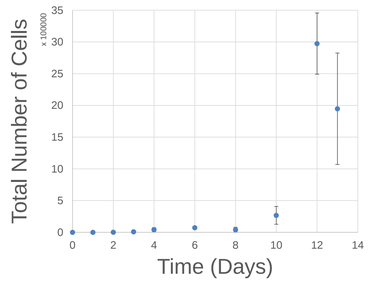
Trypan blue dye is a viable cell exclusion assay. The dye enters a perforated cell membrane in a non-viable cell. Cell counts were performed using a hemocytometer and 0.4% trypan blue dye.

http://image.slidesharecdn.com/anticanceractivitystudies-111203184417-phpapp01/95/anticancer-activity-studies-34-728.jpg?cb=1322731355

Growth and Death Curves

In order to calculate growth and death rates concurrently, an extended Monod model will be considered. This mathematical model can potentially model the βTC-6 cells, thus mimic beta cell death in diabetes.





Viable cell counts indicate a sigmoidal growth curve, while non-viable cell counts are too small to determine death rates.

βTC-6 cells were plated in a 6-well plates with 5.0 x 10^5 cells per well and the number of viable and non-viable cells were counted in triplicate up to 13 days. The media was changed only during day 1 cell counts. The mean was calculated for the graph as well as the standard deviation, which is indicated by the error bars on each graph. On day 12 and 13, cells were undergoing a death phase effecting the number of viable present by creating a acidic media, which was visible by a yellow color.

The sigmoidal growth curve was found for the βTC-6 cells. Parameters that need to be considered for growth curves is the maximum growth rate μ_m, the lag time λ, and the asymptotic value A (Zwietering et al.).

Quantifying Growth

The data points that were in the exponential phase were isolated and converted into logarithm scale by $y=\ln(N/N_o)$. Where N is the population and N_o is the initial population.
The maximum growth rate is the slope of this line

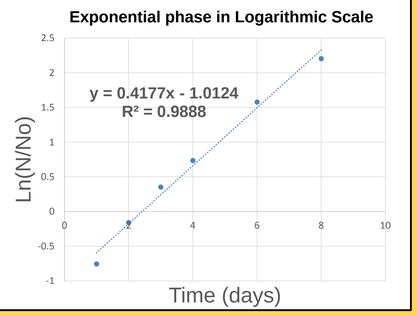
μ_m = 7.59 x 10^5 cells per day

The lag time is determine by the x-intercept of the logarithmic graph.

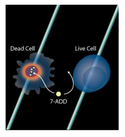
λ = 2.42 days

The asymptote is determine by $A = \ln(N_\infty / N_o)$ on the logarithmic graph,

A = 7.47 x 10^6 cells



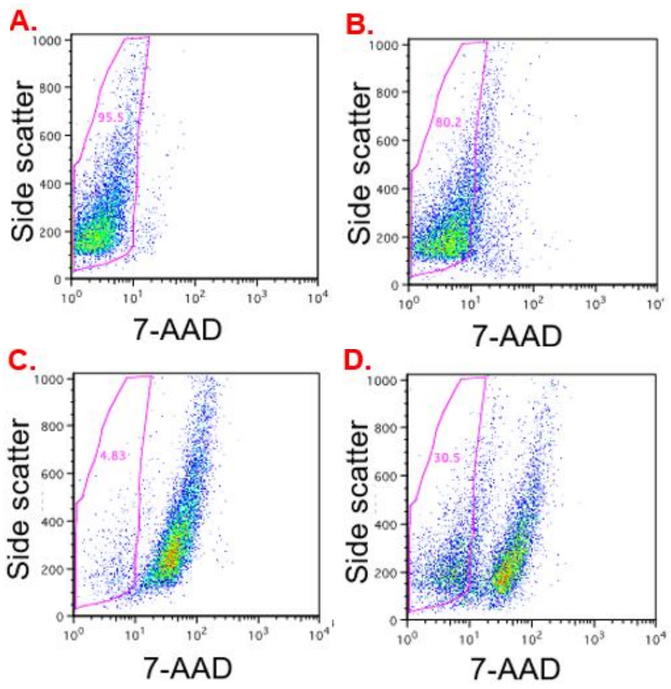
7-AAD Exclusion Assay



https://resources.mdbystems.com/images/site/fig-3.png

Flow cytometry, another exclusion assay was performed with fluorescent 7-AAD dye, which labels non-viable cells. This assay records ten thousand events leading to precise results, unlike the trypan blue exclusion assay.

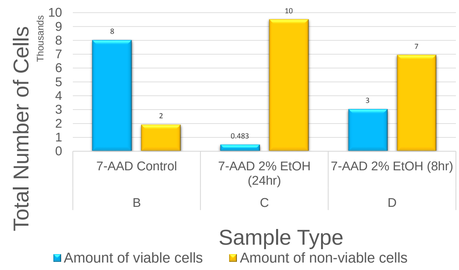
Validating Cell Counts



Flow cytometry was performed on the FACSCanto instrument. The data was verified by using a negative control with no insult and a positive control with 2% ethanol (EtOH). The 2% EtOH was added in two time points, 24 and 8 hours.

A. The unstained control, used to gate the βTC-6 cell population. While
B. The control plus 7-AAD, shows the amount of non-viable cells normally present in cell culture.
C. The 2% EtOH at 24 hours, which is completely dead. This is known since the population is outside the gate.
D. The 2% EtOH at 8 hours, which shows both population of dead and living cells.

Quantifying Flow Cytometry Results



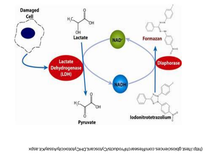
Amount of cells counted by flow cytometry with 7-AAD staining

Flow cytometry collected 10,000 events per second, this equates to 10,000 cells The amount of non-viable cells can be determined by the equation below since the gated population contains viable cells.

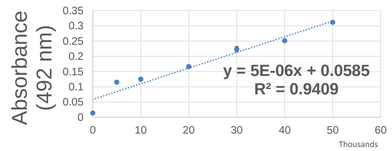
(1 - % viable) x 10,000 = Number of non-viable cells

Quantifying Lysed Cells

The dead cell population also consist of lysed cells therefore LDH also had to be considered. CytoTox 96 Non-Radioactive Cytotoxicity Assay was used to measured LDH activity in the supernatant. The reagent initiates a chemical reaction that converts tetrazolium salt into red formazan product. The amount of this product is proportionate to the amount of LDH released and thus the number of cells lysed.



Absorbance determines number of lysed cells



The titration of cells were lysed at various numbers to quantify LDH

A titration of cells was performed for the βTC-6 cells. A titration curve is necessary to verify that the kit is functioning.

Extended Monod Model

$$k_d = \frac{\{[LDH](t)/Y - [LDH](0)/Y\exp(-k_i t)\}(\mu + k_i)}{[X](t) - [X](0)\exp(-k_i t)} \qquad V_R = \frac{\mu}{\mu + k_d}$$

$$k_p = \mu + k_d$$

The Monod model does not consider the difference in toxic and growth-inhibiting influences (Renner et al.). Therefore, an extended model was considered. In this model LDH activity was quantified in the supernatant (Renner et al.). The proliferation rate k_p ,death rate k_d, and viability index V_R, are composed of three first-order kinetic equations. The inactivation rate k_i of the LDH must be determined for βTC-6 cells, in order to consider these kinetic equations. Then this potential model can be used to model beta cell death and growth simultaneously in diabetes.

Conclusions

- ❖ The trypan blue exclusion assay is only adequate for measuring viable cells to determine growth rates..
- ❖ The 7-AAD exclusion assay, precisely measures non-viable cells, unlike trypan blue exclusion assay.
- ❖ Quantifying LDH in the supernatant can determine the number of lysed cells to fully apprehend the number of dead cells.
- ❖ The extended Monod model along with the LDH measurements allows cell proliferation and death to be computed simultaneously.